

Phoenix Aerospace Engineering Platform (PAEP)

Software Requirements Specification (SRS)

Version 0.1 Alpha

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Status

Draft

Open Source Project

Executive Summary

The Phoenix Aerospace Engineering Platform (PAEP) is an open-source educational platform designed to simulate aircraft flight using real aerospace engineering principles. The platform combines flight dynamics, aerodynamic analysis, engineering visualization, and aircraft performance calculations into a single learning environment for students and researchers.

1. Project Vision

The vision of PAEP is to become a free, open-source aerospace engineering platform that enables students, educators, and researchers to explore aircraft physics through accurate simulations and interactive engineering tools.

2. Problem Statement

Most educational flight simulators are either oversimplified for educational purposes or prohibitively expensive for independent learners. This project aims to provide a free, open-source simulator that explains flight mechanics with clear physics and engineering principles.

3. Project Objectives

- Simulate aircraft motion.

- Develop mathematical models to accurately calculate aerodynamic forces including Lift, Drag, Weight, and Thrust.

- Display aircraft trajectory.

- Visualize aerodynamic forces.

- Create engineering graphs.

- Export flight reports.

- Build an educational aerospace simulation.

4. Project Scope

Included

- Aircraft equations
- Physics simulation
- Graph generation
- Aircraft performance

Not Included (Version 1)

- Multiplayer
- VR
- CFD Simulation
- Commercial Aircraft Database

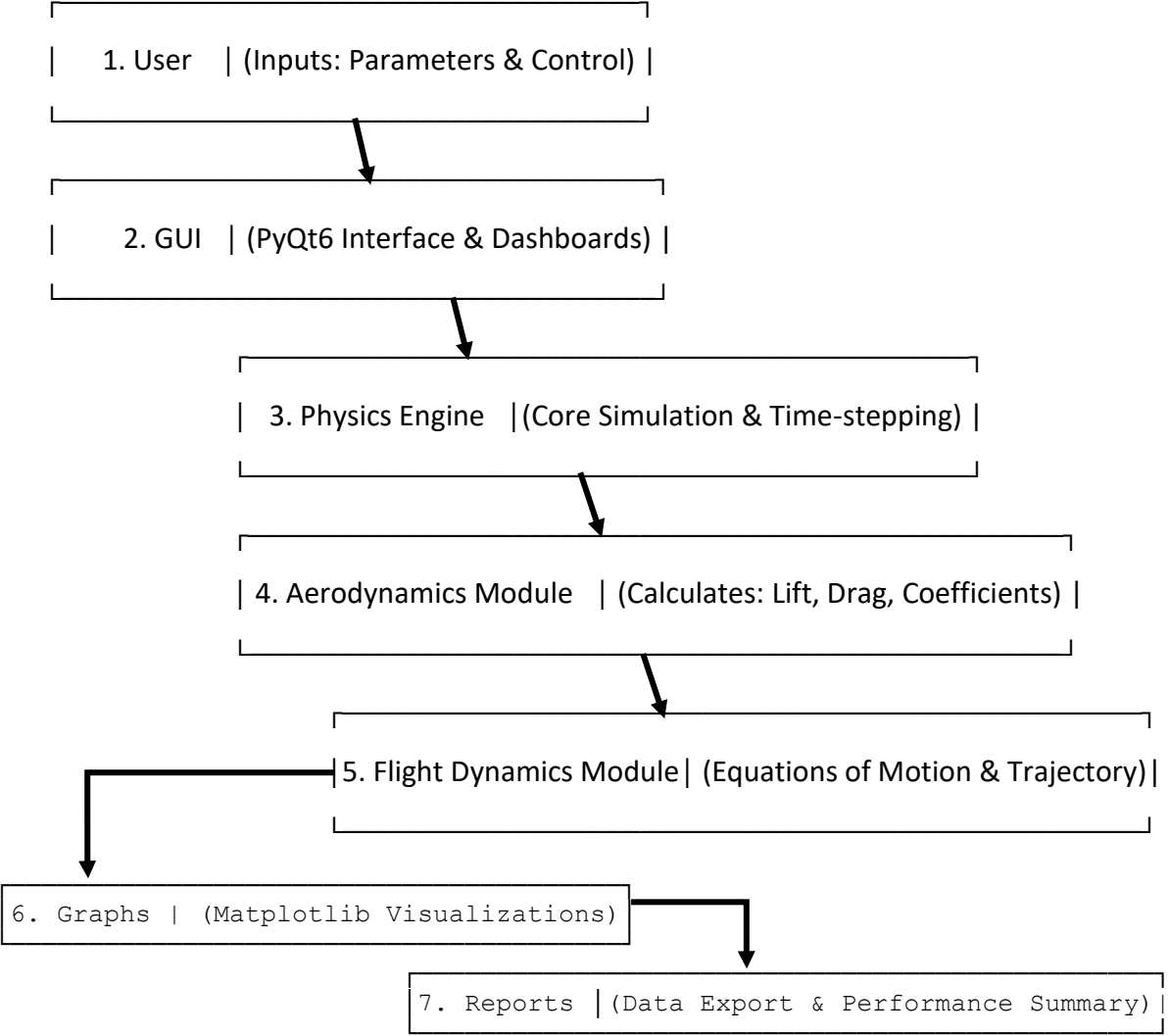
5. Functional Requirements

- The user shall enter aircraft parameters.
- The system shall calculate Lift.
- The system shall calculate Drag.
- The system shall calculate Aircraft Performance.
- The system shall display engineering graphs.
- The system shall export simulation reports.

6. Non-Functional Requirements

- Cross-platform support.
- Open-source license.
- Fast simulation performance.
- Modular architecture.
- Easy-to-use interface.
- Well-documented code.

7. System Architecture



8. Software Stack

Programming Language:

Python

Libraries:

NumPy | Matplotlib | SciPy | PyQt6

Version Control:

Git | GitHub

9. Development Roadmap

Version	Feature / Module
V1.0	Aerodynamics (Lift + Drag)
V2.0	Aircraft Motion
V3.0	GUI (User Interface)
V4.0	Wind Simulation
V5.0	Weather & Atmosphere
V6.0	AI Autopilot Module

10. Aerodynamic Equations

Lift equation: $L = \frac{1}{2} \rho V^2 S C_L$

Where:

- L = Lift Force
- ρ = Air Density
- V = Velocity
- S = Wing Area
- CL = Lift Coefficient

Variable	Meaning	Unit
L	Lift Force	Newton (N)
ρ	Air Density	kg/m ³
V	Velocity	m/s
S	Wing Area	m ²
CL	Lift Coefficient	Dimensionless

11. Target Users

- Aerospace Engineering Students
- High School STEM Students
- Physics Teachers
- Engineering Enthusiasts
- Researchers

12. Engineering Topics

- Aerodynamics
- Flight Mechanics
- Aircraft Performance
- Aircraft Stability
- Aircraft Control
- Aircraft Forces
- Atmospheric Model
- Aircraft Equations of Motion

13. Project Modules

Module	Status
Aerodynamics Engine	Planned
Flight Dynamics Engine	Planned
Aircraft Performance	Planned
Weather Module	Planned
Report Generator	Planned
GUI	Planned

14. Folder Structure

PAEP/

├─ docs/

├─ src/

├─ physics/

├─ gui/

├─ reports/

├─ assets/

├─ tests/

├─ examples/

└─ README.md

15. Future Features

- 3D Visualization
- Wind Simulation
- Weather Simulation
- Autopilot
- Different Aircraft Models
- Real Airport Database
- Mission Mode
- Aircraft Failure Simulation

16. Assumptions

The simulator assumes:

- Standard atmospheric conditions.
- No structural deformation.
- No engine failure.
- Steady wind conditions.
- Subsonic flight.

17. Long-Term Vision

This project is intended to grow into a professional educational aerospace platform. Future versions may include AI-assisted flight prediction, advanced aircraft models, real-world flight data integration, and engineering research applications.

18. Risks

Project Risks:

- Mathematical inaccuracies
- Performance issues
- Large codebase
- GUI complexity
- Physics validation

19. License

MIT License (Planned)

20. References